M. and R.
Journal of New Approaches in Educational Research (2026) 15:5
https://doi.org/10.1007/s44322-026-00052-5

Journal of New Approaches in Educational Research

ORIGINAL RESEARCH Open Access

SmartPD: a design-based research model for enhancing teacher digital competence aligned with DigCompEdu

Sathya M.[1] and Alamelu R.[1*]

*Correspondence: alamelu@mba.sastra.edu

[1] SRC, SASTRA Deemed to Be University, Kumbakonam, India

Abstract

This study introduces SmartPD, an AI-assisted, micro-learning-based professional development (PD) model designed to enhance teachers' digital competence in line with the DigCompEdu framework. Existing PD initiatives are often generic, insufficiently contextualised, and lack mechanisms for personalisation, scalability, and iterative feedback, limiting their long-term impact. To address these gaps, this research adopts a Design-Based Research (DBR) methodology, following four iterative phases: (i) contextual exploration of teachers' digital competence using the DigCompEdu Check-In tool, (ii) design and development of SmartPD modules grounded in micro-learning and AI based feedback, (iii) short-term implementation with 32 in-service teachers, and (iv) evaluation and refinement based on analytics and participant feedback. The intervention integrated H5P for interactive content, Flipgrid for peer reflections, and ChatGPT for real-time formative feedback, with learner engagement tracked through xAPI dashboards. Data were collected through pre- and post-assessments, reflective journals, and platform analytics. Results indicated statistically significant improvements in digital competence across all five DigCompEdu domains, with the most substantial gains in "Facilitating Learners' Digital Competence" and "Teaching and Learning." Teachers also reported high levels of satisfaction and confidence, while engagement data confirmed strong usability and high completion rates. Despite limitations related to sample size and duration, SmartPD demonstrates the potential of combining DBR with AI-enabled micro-learning to provide scalable, context-sensitive, and evidence-based teacher professional development, aligned with SDG 4: Quality Education.

Keywords: Digital competence, Professional development (PD), Design-based research (DBR), Artificial intelligence in education, DigCompEdu framework

1 Introduction

The increasing digitalisation of the educational process at all levels and stages of education demands defining a new paradigm for teaching teachers and assisting in their professional advancement. As education systems shift towards digital and hybrid learning, new demands are emerging in teacher professional development (TPD) programs, with an emphasis on a comprehensive understanding of digital competence, pedagogical and instructional innovation, and reflection on practice (Betancur-Chicué &

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

García-Valcárcel Muñoz-Repiso, 2023). In this case, digital competence refers to the responsible use of digital tools in planning, teaching, evaluation, and collaboration among educators, which constitutes an important component of modern and future-proof education.

Although greater awareness has been developed, there are still gaps in the current TPD offerings. Most programs are either overly generic, overly theoretical, or not relevant to classroom reality (Fernandez-Batanero et al., 2020). Such constraints tend to lead to poor engagement, insufficient transferability and a lack of long-term influence. The study by Fernandez-Batanero et al. highlights the importance of using evidence-based, iterative, and contextualised interventions that match the development stages of teachers as well as the conditions under which they work.

Recent findings suggest that teacher professional development (TPD) remains a critical concern in terms of scalability, sustainability, and contextual relevance. An example is a study by UNESCO (2023) that found that less than 40% of teachers worldwide believe they are sufficiently equipped to apply digital pedagogy in their classroom settings. On the same note, the European Commission's (DigCompEdu) monitoring of educators' digital competence (Redecker, 2017) revealed that the digital competence of educators varied considerably between regions, suggesting that skills may not be easily transferred between different areas. According to meta-reviews of TPD interventions (Philipsen et al., 2019; Fernandez-Batanero et al., 2020), even though digital competence is becoming a significant concern, the vast majority of current PD models are generic, not personalised, and include no iterative feedback or AI-based support. These inadequacies necessitate novel methods that integrate microlearning, adaptive technologies, and evidence-based models to promote effectiveness and sustainability in teacher professional development.

In order to fill this gap, there has been an increasing trend towards micro-learning, modular designs, PD and design-based innovations. Microlearning is a flexible method of learning that allows teachers to consider their time constraints and facilitates meaningful learning through brief, focused, and interactive learning experiences. According to Albion et al. (2015), the relationship between research and practice needs to be reciprocated so that such innovations become sustainable, scalable, and relevant. Their research supports sequential improvement, which implies that teachers also provide feedback on PD interventions.

As part of these emerging practices, Sharathbabu (2025) emphasise the need for adaptive and AI-enabled platforms that help teachers empower their autonomy and promote digital literacy throughout the process. Their study reveals that personalised and collaborative learning spaces significantly enhance the competence and confidence of teachers when peer networks and real-time digital feedback systems are promoted.

Based on these ideas, this work develops the SmartPD model, a structured, microlearning-based professional development intervention grounded in Design-Based Research (DBR) for steep professional development. SmartPD aims to cover both pedagogical and technological aspects of digital competence, including the use of H5P (to create interactive materials), Flipgrid (to facilitate collaboration on reflections), and ChatGPT (to introduce AI-based feedback and personalisation). The combination of these tools into a DBR framework should enable the model to build a dynamic,

context-sensitive learning ecosystem that changes and evolves in response to the input provided by participants and in real-time based on real-time analytics. The European Commission's DigCompEdu framework is also a guiding principle of the intervention, ensuring that the development of digital skills aligns with established and validated international standards.

Aligned with the goals of SDG 4 (Quality Education), this study aims to contribute to the global dialogue on equitable and practical teacher training. Specifically, the research pursues the following objectives:

(i) To design and implement the SmartPD model grounded in micro-learning and AI tools,
(ii) To evaluate its effectiveness in enhancing teachers' digital competencies, and
(iii) To understand teachers' experiences and engagement during the intervention.

To assess these objectives empirically, the study employs a quasi-experimental design with pre- and post-intervention measures, enabling the evaluation of changes in teacher competence and engagement. The SmartPD intervention combines micro-learning modules, interactive tools, and AI-driven feedback mechanisms to create a dynamic and responsive learning experience. By measuring both cognitive and affective outcomes—such as digital skills acquisition, instructional decision-making, and user satisfaction—the study seeks to provide robust evidence of the model's practical value.

To operationalise the above research objective and provide the study with a narrow analytical focus, the following research questions are used to organise the study. These questions put the objectives in the form of empirically testable questions that will direct both the quantitative and qualitative analyses.

The research questions necessitate a mixed-methods approach to answer them. The quantitative data from the pre- and post-intervention DigCompEdu tests, along with correlational analyses, are used to investigate RQ1 and RQ3. Both quantitative indicators of engagement and satisfaction, as well as qualitative reflective data from respondents, help to investigate RQ2.

RQ1. How does the SmartPD intervention enhance the digital competence of teachers, as measured by pre- and post-assessed digital competence scores using DigCompEdu?

RQ2. To what extent do teachers evaluate their participation in the SmartPD intervention and their satisfaction with it, using both quantitative measures and qualitative data?

RQ3. What are the correlations between the post-intervention decision-making and digital-feedback skills of teachers and other DigCompEdu competency areas, and how far do these areas forecast change in such skills, controlling for the baseline levels?

2 Literature review

2.1 Digital competence in teacher education

Betancur-Chicue and Garcia-Valcarcel Muizon-Repiso (2023) emphasise the need to develop digital competence in teachers in specific areas related to feedback and decision-making, as the development of these competencies often lags. To address this, they suggest introducing microlearning to enhance their skills. According to Ajani (2024), the professional development of teachers should focus on increasing the digitalisation of the classroom through aligning technological, pedagogical, and content knowledge. Jo Tondeur et al. (2016) review the obstacles to integrating ICT and reveal that existing TPD programs have structural and pedagogical barriers. The case study given by Cervera and Cantabrana (2015) suggests that teacher digital competence is associated with the enhancement of overall school quality.

In addition to DigCompEdu, several international models exist that facilitate the growth of digital competence among teachers. The UNESCO ICT Competency Framework for Teachers (UNESCO, 2018) focuses on the integration of ICT as a means to acquire knowledge, enhance pedagogy, and develop professional skills; however, it has been criticised as being too generic and challenging to apply contextually to particular learning environments (Nguyen & Habók, 2023). According to the ISTE Standards for Educators (ISTE, 2022), the focus is on empowering learners, fostering teamwork, and promoting digital citizenship; however, the orientation towards North America may limit its international competence (Koh & Chai, 2016). DigCompEdu offers a more detailed, practice-oriented model that is data-driven explicitly for European and international teacher settings, as compared to these models (Redecker, 2017). The six competence areas (e.g., digital resources, assessment, and empowering learners) enable direct mapping to classroom practice. Therefore, they can be empirically tested with the help of professional development models, such as SmartPD.

2.2 Gaps in existing professional development models

Fernandez-Batanero et al. (2020) conducted a systematic review of PD programs for teachers, showing their inability to adapt, be scalable, and relevant to practice changes in the digital world. Cross et al. (2022) demonstrate the problem of scaling practice-based models of PD in countries of the Global South, where the lack of resources is evident in technology and feedback systems. Napitupulu et al. (2025) presents the case of digital transformation of TPD on a global scale, highlighting problems of sustainability, usability, and appropriateness in relation to institutional policies. Fakhar et al. (2024) suggest that there is a necessity to have AI-powered PD frameworks that will be continuous, context-sensitive, and customised. The study by Langset et al. (2018) emphasises the effectiveness of collaborative learning experiences in having a positive impact on digital literacy in the context of higher education. Wuryaningsih et al. (2018) review online models of PD and suggest more interactive, practice-oriented practices and socially driven program designs.

One of the challenges that has remained complicated in TPD models is their scalability and sustainability. Whereas digitisation is becoming a part of PD programs in the Global North, research in the Global South indicates a high rate of infrastructural and policy-based barriers to adaptation (Cross et al., 2022; Syifa et al., 2024). In Indonesia

and sub-Saharan Africa, for example, teacher digital upskilling programs often fail to maintain their course due to the lack of institutional support and access to digital infrastructure (Napitupulu et al., 2025; Wuryaningsih et al., 2018). In addition, the majority of programs focus on generic training in ICT, as opposed to context-sensitive training in digital pedagogy, which makes classrooms less transferable. SmartPD is a direct response to those deficits, as it integrates AI-powered personalisation, microlearning modules, and contextual adaptation based on DBR, potentially providing a viable and sustainable alternative.

2.3 Theoretical frameworks & design-based research (DBR) in professional development

Design-Based Research (DBR) has become a practical methodological and theoretical approach to organising teacher professional development (PD), particularly in settings where digital competencies are rapidly evolving. It encourages progressive improvement, exemplification of workflows, and field-tested usage of education-based operations. Fowler and Leonard (2021) emphasise that DBR presents a sustainable professional development model because it makes a conscious effort to transform the minds of professionals through reflective, iterative, and problem-specific design steps. In their work, they focused more on the fact that DBR aligns with the context of real-life classroom dynamics, which in turn supports innovation in the use of digital tools. Likewise, Zinger et al. (2017) indicated that a DBR framework resulted in a significant improvement in teacher knowledge and the quality of PD design, due to continuous feedback loops that contribute to the improved achievement of learning outcomes and digital integration.

Wang et al. (2014) used DBR to enhance the utilisation of ICT as cognitive tools among teachers. They emphasised how iterations of refinement, based on authentic classroom issues, helped teachers change their approach to technology integration from superficial to deep use. This literature is substantial in favour of using DBR in SmartPD since learning modules are developed through iterative piloting and teacher feedback. Albion et al. (2015) supplement this with the advocacy of a mutually beneficial relationship between research and practice in ICT-PD, emphasising co-design and shared ownership. Tregubova et al. (2020) further support the belief that prototyping progressively and engaging stakeholders is an important concept in designing PD frameworks in digital settings. Syifa et al. (2024) put the rationale behind the application of DBR in the low-resource context, declaring that it must be linked to sustainable PD, which demands alignment across the technology, infrastructure, and teacher-building capability aspects.

In comparison to quasi-experimental designs, DBR involves a concurrent cycle of design, enactment, and evaluation, thereby being inherently iterative, context-sensitive, and collaborative (McKenney & Reeves, 2018; Fowler & Leonard, 2021). This aspect renders DBR especially suitable for investigating intricate educational interventions in the interplay of teacher agency, technology, and institutional settings. In SmartPD, DBR enabled a gradual process of prototyping: the design was based on the needs analysis, the daily prompts and AI responses were considered micro-iterations, and the teacher reflections prompted the improvement. Such alignment ensures that the design is not fixed but adapts to the realities of the classroom by connecting research with practice and enhancing the rigour of the study methodology.

2.4 Micro learning, AI, and analytics in PD

Microlearning has a bright future in developing digital competence. The research by Betancur-Gu Cherique-Chiu (2023) confirms the validity of a microlearning model of DigCompEdu through digital feedback. The study conducted by Alcaraz-Dominguez and Barberà (2025) assessed the needs of educators in their training in Spain, focusing on how the concept of microlearning, based on DigCompEdu-aligned micro-units, is an effective way to address competence deficiencies in the area of digital teaching. This is evidenced by the inclusion of micro learning strategies in scalable, need-based models of professional development. Fakhar et al. (2024) describe the review of AI-based models of continuous PD and recommend real-time feedback and on-demand learning during the upskilling process of teachers. According to Sharathbabu (2025), learning environments and collaboration with an AI-mediated and adaptive learning environment are effective ways to enhance teacher digital literacy.

According to the recent systematic reviews, AI-based adaptive feedback and learning analytics dashboards are transforming the professional development of teachers ((Banihashem et al., 2025; Tan et al., 2024). Adaptive systems will be able to customise the pacing, detect competence gaps and deliver real-time formative feedback (Wang & Han, 2022). In contrast, analytics will provide institutions with the data to monitor engagement and performance (Viberg et al., 2018). However, concerns remain about equity, ethical use, and the digital divide. Research also warns that AI-based PD can exacerbate inequalities in the absence of adequate infrastructure or digital literacy (Holmes et al., 2023). In this way, regardless of the potential of AI and analytics, their application should be pedagogically based and morally accountable. SmartPD integrates these understandings through the inclusion of real-time AI-based feedback and xAPI-driven analytics, as well as flexibility to various teaching situations.

2.5 Research gap

Recent studies have affirmed the possibility of utilising microlearning in meeting particular skills needed to become digitally competent. Betancur-Chicué (2023) tested a DigCompEdu-aligned microlearning model on feedback-related competencies and demonstrated its effectiveness in the form of short, contextual learning bursts. Continuing this line of thought, an empirical experiment in Spain by Alcaraz-Dominguez and Barbera (2025) demonstrated that programmable microlearning modules were highly aligned with the training needs identified by educators, specifically in the area of digital pedagogy. These results confirm the microlearning design principles of the SmartPD model, which assumes the authenticity of competency matching. Moreover, Fakhar et al. (2024) proposed an AI-inspired professional development model that focuses on the concept of real-time flexibility, and Sharathbabu (2025) demonstrated the usefulness of AI-mediated spaces in developing collaborative learning contexts. Taken together, these studies emphasise the necessity of incorporating microlearning, feedback analytics, and artificial intelligence into a single system of professional development. In order to guarantee the conceptual clarity and measurement validity, the most critical constructs guiding the research questions, i.e., engagement, satisfaction, instructional decision-making, and digital feedback skills, are operationally defined in the following section.

2.6 Research design and methodology

Although digital competence frameworks and TPD models have advanced, three apparent gaps can be identified. First, although DigCompEdu has a systematic structure, there is a lack of studies that have operationalised it with AI-based micro-learning interventions that directly correlate competencies with practice. Second, current PD studies often employ a quasi-experimental design, and rarely utilise iterative cycles of DBRs that would enable adaptation to the context and continuous improvement (Philipsen et al., 2019; Zinger et al., 2017). Third, despite the increased research in AI and analytics in the educational field, there is a lack of information on how it facilitates the teacher decision-making process and feedback-related competencies, which are fundamental to effective digital pedagogy (Betancur-Chicué & García-Valcárcel, 2023; Fakhar et al., 2024). Combining DBR methodology and AI-driven microlearning in line with DigCompEdu, this study fills the research gap. It demonstrates that a scalable, context-sensitive professional development model can be used to promote not only teacher practice but also global commitments to SDG 4.

2.7 Hypothesis formulation

Based on the research objectives and questions, the following hypotheses were formulated to guide the quantitative analysis and test the directional relationships between critical variables that were expected. These hypotheses provide a guided framework for testing the expected impacts of the SmartPD intervention.

H1: Participants will demonstrate a significant improvement in digital competence post-intervention.
H2: Teachers will report high engagement and satisfaction with the SmartPD model.
H3: The integration of micro-learning and AI tools will support decision-making and digital feedback skills.

2.8 Research design framework

In Fig. 1, the steps followed by the phased approach of the SmartPD intervention study, which is designed based on the Design-Based Research (DBR) methodology, are provided. It describes a systematic development and implementation process that follows a cycle, running from needs evaluation to needs assessment.

Design-Based Research (DBR) has been chosen as the methodological framework because it is an explicitly conceptualised method of linking theoretical constructs to

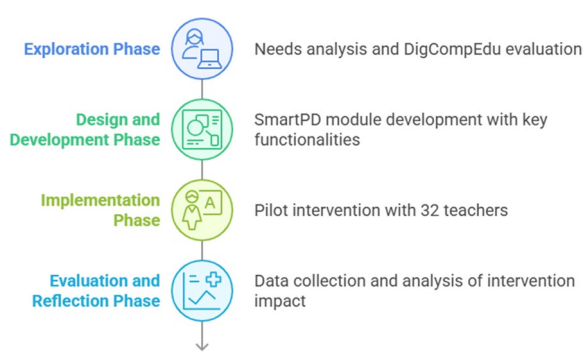

Fig. 1 Design-based research cycle. Source: Author (s)

classroom practice through cyclic exploration, design, enactment, and analysis (McKenney & Reeves, 2018; Wang & Hannafin, 2005). Compared to single quasi-experimental interventions, DBR encourages the gradual development of innovations in real-world contexts and focuses on working with practitioners, making it especially applicable to the research of teacher professional development (Fowler & Leonard, 2021). The application of DBR in the study meant that SmartPD was not a rigid program, but a dynamic design that constantly evolved with the help of teacher feedback, AI-generated responses, and analytics to inform its implementation. Micro-iterations were present even within a two-week cycle, achieved through daily adjustments of the prompts, refinements of AI-based feedback, and adjustments to the collaborative activities based on participant engagement.

To strengthen alignment between research goals and design, the study objectives were mapped directly onto the DBR phases, as shown in Table 1.

The 32 in-service teachers' sample was also selected based on purposive sampling to ensure that it represented a mixed field of disciplines and the degree to which they had been exposed to ICT in the past. This purposive sampling provided the necessary ecological validity because it represented a wide range of teaching conditions, which aligns with the focus of DBR on context-specific observations rather than generalisation (Zinger et al., 2017).

a. Exploration Phase
   A needs analysis was used as the starting point of the research to diagnose digital competency levels among in-service teachers participating in the study. An adapted DigCompEdu Check-In tool was used to evaluate the strengths and weaknesses of the participants in the six key competency areas of the framework, as they existed at the moment of the measurement. The observations made during this diagnostic point were used to inform the design orientation of the SmartPD module.
b. Design and Development Phase
   The SmartPD module was discussed and developed accordingly, incorporating functionalities that addressed the gaps identified during the exploration phase. These instructional strategies focus on microlearning, individualisation, collaborative learning, and real-time feedback. Technological tools incorporated into the platform were:

Table 1 Alignment of research objectives, DBR phases, and methods

Research objective	DBR phase	Methods/activities
(i) To design and implement the SmartPD model grounded in microlearning and AI tools	Exploration → Design & Development	Needs analysis using DigCompEdu Check-In; design of SmartPD modules; tool integration (H5P, Flipgrid, ChatGPT)
(ii) To evaluate its effectiveness in enhancing teachers' digital competencies	Implementation → Evaluation	Pre–post DigCompEdu assessments; paired-sample t-tests; effect size calculations; xAPI analytics
(iii) To understand teachers' experiences and engagement during the intervention	Implementation → Evaluation → Redesign	Reflective journals; AI feedback logs; optional focus group interviews; thematic coding

Source: Author(s)

- H5P: for interactive and engaging content delivery
- Flipgrid: for asynchronous video-based collaboration
- ChatGPT: to simulate pedagogical dialogues and provide formative feedback
- xAPI dashboards: to track learner engagement and performance.

The content was mapped to the DigCompEdu competence areas, with scenarios and tasks aligned to participants' teaching contexts.

c. Implementation Phase
The pilot intervention was conducted over two weeks with a purposive sample of 32 in-service teachers from various disciplines. The SmartPD platform enabled both asynchronous and synchronous learning interactions, allowing teachers to fulfil modular assignments that revolved around specific digital competencies. Nevertheless, although the timeline was allocated to be shorter, the intervention has depth, incorporating daily prompts, AI-based feedback, and peer reflections within the framework of collaborative learning.

d. Evaluation and Reflection Phase
To measure the impact of the intervention, multiple forms of data were collected and triangulated:

- Pre- and post-intervention surveys using the DigCompEdu tool to assess growth in digital competence
- The SmartPD platform utilised xAPI (Experience API) to track teachers' interactions with learning activities, including task completion, video views, and discussion participation. This data provided insights into their engagement and learning behaviour.
- Open-ended reflection forms and optional focus group interviews to gather qualitative feedback
- AI-generated feedback logs were also reviewed for pedagogical relevance and teacher satisfaction.

Analysis of the data was conducted using both descriptive statistics and inferential statistics for the quantitative responses, and thematic coding for the qualitative responses. The results of this step were used to guide subsequent modifications to the SmartPD design and to provide evidence of its success.

2.9 Ethics and informed consent

The Institutional Review Board of the host institution approved this study. Informed consent was obtained from all participants before they participated in the study. They were fully aware of the study's purpose. They assured that the information they provided would not be disclosed to outside parties and that the collected data would only be used for research purposes.

2.10 Participants

The sample consisted of 32 in-service teachers across disciplines, with varying levels of ICT exposure. Participants were selected through purposive sampling based on their expressed interest in digital pedagogy training.

2.11 Instruments and measurement

2.11.1 DigCompEdu check-in instrument

The research design employed a multi-instrument data collection plan, tailored to the research design and the evaluation purpose. Table 2 presents the data collection instruments aligned with each phase of the SmartPD intervention, explaining their purpose and application in accordance with the Design-Based Research model. The DigCompEdu Check-In tool (Redecker, 2017) was localised by making some minor linguistic simplifications without altering the original construct's meaning. The instrument did not change structure or concepts. To assess the clarity and relevance of the adapted version, an expert review was conducted, and internal consistency (Cronbach's α) values across all domains indicated that the version is acceptable for use in this study.

2.11.2 Reliability and sample items

The internal consistency of the DigCompEdu Self-Assessment instrument was tested to ensure that the competency domains used in the research were reliable. The Cronbach's alpha (α) of each of the five domains was estimated, and all coefficients had acceptable to good reliability (Cronbach's alpha values indicated acceptable to good reliability for all five DigCompEdu domains ($\alpha = 0.744–0.852$), confirming that each domain consistently measured the intended construct). These values affirm that the items in every domain assessed the intended constructs reliably.

Sample questions in each of the competency areas are provided below to give a better understanding of how the DigCompEdu domains will be operationalised. The following items indicate the kind of activities and self-assessment judgments that teachers made when filling out the pre-assessment and post-assessment:

Table 2 Data collection tools, purpose and its application

Tool	Purpose	Application phase	Reference
DigCompEdu Self-Assessment (validated)	Measure pre- and post-intervention digital competencies	Exploration, Evaluation	(Párraga et al., 2022)
xAPI Dashboards	Track user engagement, activity completion, learning behaviors	Implementation, Evaluation	(ELM Learning, 2024)
Open-ended Reflections	Collect participant feedback on usability and learning experience	Implementation, Evaluation	(Schürmann et al., 2025)
Focus Group Interviews (optional)	Gather deeper qualitative insights on tool use and satisfaction	Evaluation	(Frasier et al., 1997)
AI Feedback Logs	Examine AI-generated support for pedagogical relevance	Implementation, Evaluation	(Banihashem et al., 2025)

Source: Author

Digital Resources.

"I can create and modify digital resources to support my instructional goals."

Teaching and learning.

"I integrate digital tools into lesson activities to enhance student engagement."

Assessment.

"I use digital platforms to assess student learning and provide formative feedback."

Empowering Learners.

"I adapt digital activities to meet diverse learner needs and promote autonomy."

Enabling the Digital Competence of the Learners.

"I guide learners in developing responsible and effective use of digital tools."

These items demonstrate the level of competency expectations outlined in the DigCompEdu framework and provide the context in which the measured shifts in digital competence are interpreted.

2.11.3 Mapping to DigCompEdu framework

The intervention prioritises instructional-facing DigCompEdu domains, as these directly support classroom-level digital teaching competencies. Professional engagement, while important, lies beyond the core instructional focus of the SmartPD framework. Each module was aligned with specific DigCompEdu areas:

- Digital Resources,
- Teaching and learning,
- Assessment,
- Empowering Learners,
- Facilitating Learners' Digital Competence.

Figure 2 presents the DigCompEdu framework adapted from Redecker (2017), outlining six key areas of educators' and learners' digital competence, serving as the foundation for SmartPD's competency mapping.

3 SmartPD intervention design and components

The SmartPD model was conceptualised as an intervention framework, a technology-driven model of professional development designed to develop teachers' digital competence. The model incorporates elements of modular micro-learning, real-time feedback, tracking, and collaborative learning, aligning with the DigCompEdu framework.

Figure 3 shows essential elements incorporated into the SmartPD intervention design. It highlights the collaboration between digital competence mapping, modular micro-learning, AI-based feedback, collaborative learning and xAPI analytics.

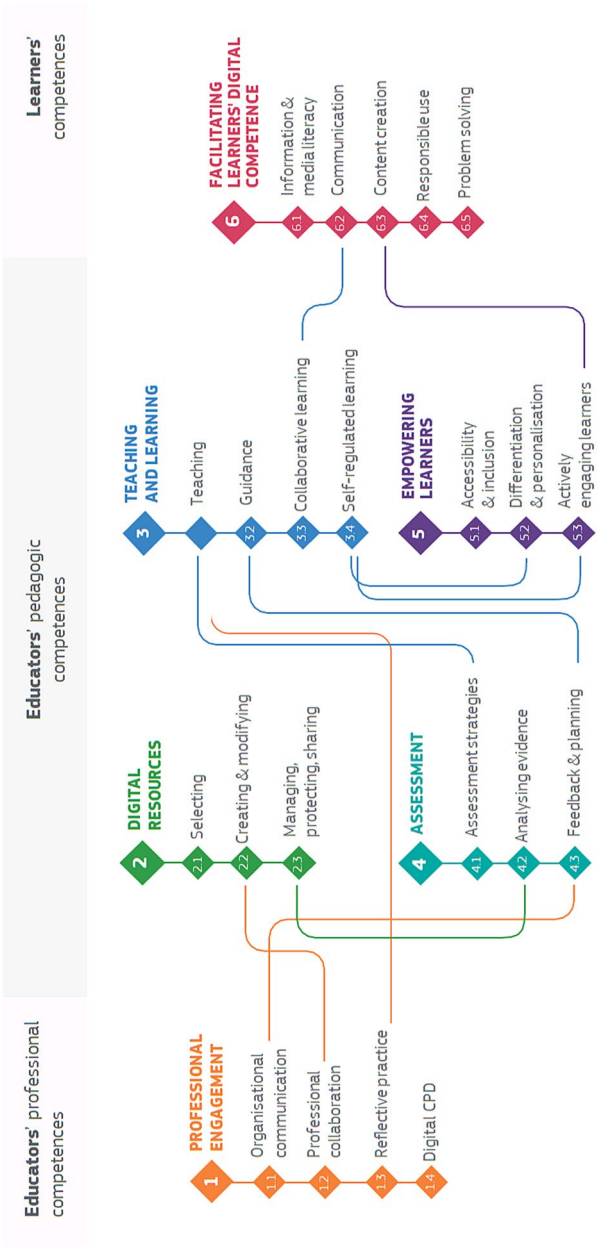

Fig. 2 The DigCompEdu framework. Source: Redecker, C. (2017)

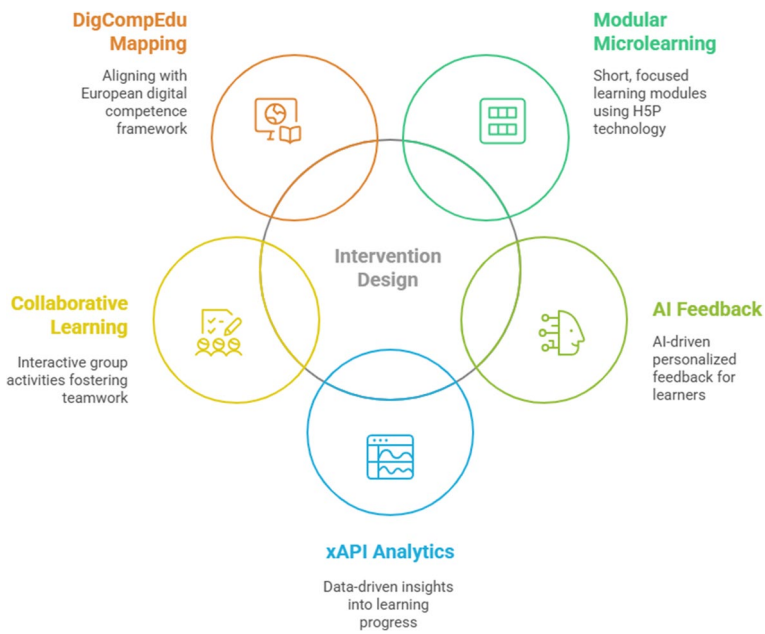

Fig. 3 SmartPD intervention components. Source: Author (s)

3.1 Modular micro-learning with H5P

The training was delivered through H5P-based interactive micro-lessons, organised into four modules over two weeks. Each module focused on a specific core area of digital competence: creating and modifying digital resources, engaging learners with technology, assessing digitally, and fostering digital collaboration. Designed for asynchronous access, these micro-lessons provided teachers with the flexibility to engage with the content at their own pace while maintaining pedagogical integrity and instructional coherence.

3.2 AI-powered feedback mechanism

The SmartPD intervention combined AI-based tools to provide automated, formative feedback on open responses and quiz tasks. Self-regulated learning was made possible with instantaneous suggestions, hints, and summaries of their performance as provided by the teachers.

3.3 xAPI-based learning analytics dashboard

To monitor user engagement and learning behaviour, the platform utilised xAPI (Experience API) to track key metrics, including time spent per module, completion rate, frequency of discussion participation, and patterns of re-attempts and revision behaviours. These data points provided detailed insights into learner activity and interaction with the SmartPD modules. The tracked metrics were visualised through an interactive dashboard accessible to both facilitators and participants, enabling real-time monitoring and informed pedagogical decision-making.

3.4 Collaborative learning elements

The model incorporated the use of discussion forums, peer review, and group activities to promote learning among teachers. The participants reflected on, evaluated, and engaged in well-organised collaborative problem-solving activities, focusing on the digital work of their peers.

3.4.1 Definitions of key constructs

To ensure conceptual clarity and measurement validity, the subsequent key constructs used in the research questions and hypotheses are defined in accordance with the available literature.

Engagement in this study refers to how teachers participate in SmartPD activities and the behaviours, emotions, and cognition of teachers during these activities, such as their participation in micro-lessons, discussions, and reflective activities. The engagement is conceptualised in accordance with Fredricks et al. (2004), who define engagement as a multidimensional construct that captures longer-term involvement, effort, and task-oriented investment in learning processes.

Satisfaction refers to the teachers' perception of the overall quality, relevance, and usability of the SmartPD intervention. It represents the affective assessment of what users have learned, including perceived value, clarity of the activities, and ease of use of the digital tools, in line with the framework of technology-enhanced learning satisfaction suggested by Kara M., (2020).

Instructional decision-making is also defined as the skills of teachers to choose, modify, and utilise the appropriate digital tools and pedagogical strategies in line with the needs of learners and learning goals. It also conforms to the theory developed by Shavelson (1973), which conceptualises decision-making as the process by which teachers consider information and make informed choices in implementing instruction in dynamic learning situations.

Digital feedback skills refer to the ability of teachers to deliver prompt, practical, and pedagogical feedback via digital platforms. These involve the development of online formative assessment, marking of the learner's answers and provision of online learner guidance. The definition aligns with that of Henderson et al. (2019), which also emphasises that technology-mediated feedback can help learners improve.

3.5 Data analysis

To interpret the data in accordance with the research questions, a mixed-methods approach combining quantitative and qualitative analysis was employed. Mean, standard deviation, skewness, and kurtosis were calculated as descriptive statistics for the individual DigCompEdu domain. Shapiro–Wilk tests were then performed to determine whether the difference scores were normally distributed, and paired-sample t-tests were used accordingly. Paired t-tests were subsequently applied to compare the pre- and post-intervention scores, and Cohen's d values were generated to estimate the effect sizes.

The Pearson correlation coefficients were calculated to test the relationship between post-intervention competencies. Open-ended reflections and voluntary focus group feedback, provided as qualitative data, were analysed through reflexive thematic analysis,

based on the six-phase process suggested by Braun and Clarke (2006). The analysis procedure included familiarisation with data, inductive coding, generation of themes, review, and refinement. The coding involved two researchers, and the refinements of meanings and boundaries of the themes were carried out through the use of iterative discussions, as opposed to establishing inter-coder reliability. The resulting themes were used to supplement the quantitative results and provide further insight into teacher engagement and satisfaction.

Although H3 was presented as a directional hypothesis, correlation analysis was used as an exploratory method to examine the relationships between instructional decision-making, digital feedback skills, and other domains of DigCompEdu following the intervention. This approach was chosen due to the small sample size and the observed relationships. The results are, therefore, taken as indicative and not causal.

4 Results

This section presents the findings that align with the set of three research questions. The collected data were the result of pre- and post-assessment surveys, observation notes, and participant comments, which allowed for drawing both quantitative and qualitative conclusions. The analysis is based on testing whether the SmartPD model is efficacious in improving digital competence, user satisfaction, and engagement in instruction delivery among teachers. Both descriptive and inferential statistical methods were used to provide a sound interpretation of the results. The substantial improvements were observed in all measured dimensions, further supporting the possibility of AI-integrated micro-learning strategies in teacher professional development (Betancur-Chicué & García-Valcárcel, 2023; Alcaraz-Domínguez & Barberà, 2025).

4.1 Descriptive statistics and assumption checks

The pre-post differences in all five domains of DigCompEdu were positive, with the parametric test assumption being met, except for the Empowering Learners domain, which underwent a robustness test using the Wilcoxon test. These exploratory findings served to substantiate the measurement of steady progress in the domains. They formed the foundation on which hypothesis testing within the framework of Objective 1 was to take place. In the case of the Wilcoxon signed-rank test of the Empowering Learners domain, the effect size (r) was determined according to the standard value of the non-parametric test $r = Z/\sqrt{N}$.

Figure 4 shows the mean difference (Post − Pre). Difference scores were run through Shapiro–Wilk tests. Empowering Learners was not normal ($W = 0.923$, $p = .032$), and thus a Wilcoxon signed-rank test was also provided. Paired t-tests are two-tailed ($df = 29$). The d (paired) of Cohen = mean difference/SD of the difference scores.

4.2 Statistical effects of the SmartPD intervention

The primary objective was to evaluate the effectiveness of the SmartPD intervention when implemented in a real educational setting. The findings of the paired-samples t-test, as shown in Table 3, indicate that all five domains of DigCompEdu significantly improved after the two-week intervention ($p < .001$). To obtain a consistent interpretation, paired-sample t-tests were calculated using the difference score (Post—Pre). This

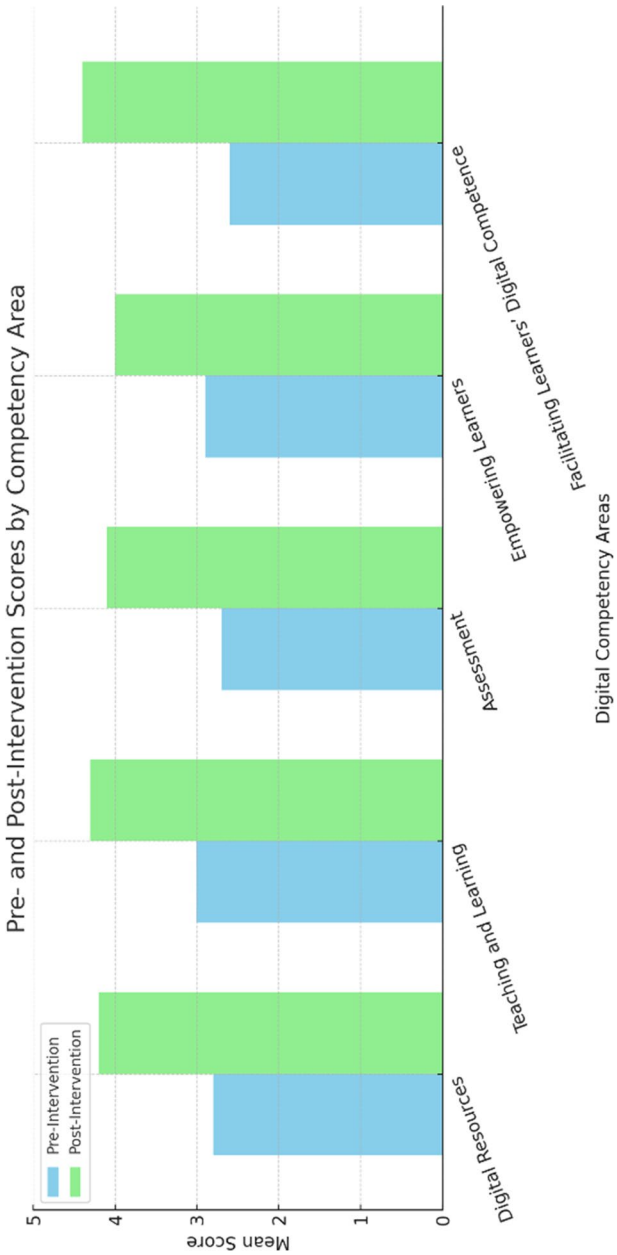

Fig. 4 Pre and post comparison. Source: Author(s)

Table 3 Paired-sample t-test results of SmartPD intervention

Competency area	t	df	p
Digital Resources	−14.799	31	<.001
Teaching and Learning	−12.457	31	<.001
Assessment	−12.911	31	<.001
Empowering Learners	−10.825	31	<.001
Facilitating Learners' Digital Competence	−23.109	31	<.001

Source: Author

means that positive differences in means imply positive t-values and improvement in the DigCompEdu domains. These gains suggest that the design of SmartPD, comprising modular lessons of H5P microlearning, AI-generated formative feedback, and xAPI-based analytics, enabled educators to make meaningful progress in specific digital skills. The high effect sizes in both Facilitating Learners' Digital Competence and Teaching and Learning are indicative of the fact that the combination of self-directed micro-lessons with automated instructions induced more substantial engagement and solidified learning through practice. This is consistent with the previous studies that indicate that iterative technology-enhanced PD models based on authentic tasks can facilitate a significant change in competency (Wang et al., 2014). Provisional comparisons with previous models of PD, such as the blended digital competence program developed by Elsayary (2023), suggest that SmartPD may offer even better returns due to the synergy it forms between micro-learning granularity and AI-mediated feedback loops.

Mean difference = Post − Pre. Paired t-tests are two-tailed. Cohen's d (paired) = mean difference/SD of difference scores. Positive t-values indicate positive change. To determine the practical significance of these improvements, Cohen's d (paired) values were calculated in each domain. The magnitude of all the effect sizes was even larger than the value of a significant effect ($d \geq 0.80$), indicating that the observed gains were not only statistically significant but also educational. The greatest impacts were also observed in Facilitating Learners' Digital Competence ($d = 5.963$), Teaching and Learning ($d = 5.565$), followed by Digital Resources ($d = 4.640$), Assessment ($d = 4.700$), and Empowering Learners ($d = 3.344$). These are magnitudes that easily surpass standard PD thresholds, as reported in previous literature, with the magnitude of effects typically falling within the range of moderate to large (e.g., $d = 0.65–0.95$ in Elsayary, 2023). This means that the combination of AI-based feedback, modular micro-learning, and cyclic practice in SmartPD resulted in remarkably high levels of competence improvement in a brief intervention time.

4.3 Results for RQ1: digital competence development

This subsection discusses RQ1, which focuses on evaluating the digital competence of teachers after the SmartPD intervention. The results demonstrate clear post-intervention improvements across all domains, as all domains improved, as expected by AI-driven PD frameworks. The statistical evidence of improvement was determined under Objective 1; however, the current subsection of this paper puts these results into the context of H1. The scale and regularity of the increase support the notion that micro-learning

Pre- and Post-Comparison Across Domains

Domain	Pre Mean	Pre SD	Pre Skewness	Pre Kurtosis	Post Mean	Post SD	Post Skewness	Post Kurtosis	Mean Diff (Post−Pre)	SD Diff	t (df=29)	p	Shapiro-W	Shapiro-p	Cohen_d_paired	Wilcoxon V	Wilcoxon p
Digital Resources	2.677	0.5	0.149	2.802	3.756	0.567	−0.279	2.816	1.08	0.233	25.412	< .001	0.9835	0.9098	4.64	—	—
Teaching and Learning	2.858	0.6	−0.539	3.621	4.122	0.605	−0.772	3.489	1.264	0.227	30.479	< .001	0.9477	0.1471	5.565	—	—
Assessment	2.865	0.59	0.285	2.241	3.992	0.578	−0.134	2.376	1.127	0.24	25.746	< .001	0.9562	0.2476	4.7	—	—
Empowering Learners	2.844	0.63	1.015	6.459	3.951	0.504	−0.225	2.92	1.107	0.331	18.318	< .001	0.923	0.032	3.344	0	< .001
Facilitating Learners' Digital Competence	2.668	0.58	−0.180	3.502	3.966	0.608	−0.392	3.803	1.298	0.218	32.662	< .001	0.9914	0.9964	5.963	—	—

Fig. 5 Pre-Post intervention mean score. Source: Redecker, C. (2017)

formats with the addition of AI-based personalisation and immediate feedback can reinforce digital pedagogic practices (Betancur-Chicué & García-Valcárcel, 2023; Fowler & Leonard, 2021). The course and magnitude of the change also support the theoretical supposition that feedback responsiveness and modular practice cycles provide measurable learning benefits.

The bar chart in Fig. 5 presents a comparison of mean scores before and after the SmartPD intervention across five digital competency domains. All areas show marked improvement, with "Facilitating Learners' Digital Competence" and "Teaching and Learning" exhibiting the highest gains. This demonstrates the positive impact of the SmartPD model on strengthening educators' digital proficiency.

Thus, this empirical evidence validates the effectiveness of the SmartPD model and its alignment with the goals of SDG 4 (Quality Education), particularly in promoting equitable access to high-quality professional development in digital education. These results indicate that SmartPD can be used to improve the digital skills of teachers, providing good empirical evidence to support RQ1.

4.4 Results for RQ3: relationships between competency domains

This analysis was conducted using the Pearson correlation matrix (Table 4), which enabled the evaluation of the relationships between the different domains of digital competence developed following the SmartPD intervention. This discussion addressed RQ3, which concluded with the detection of improvement in a coordinated or independent manner across the areas of DigCompEdu. Empowerment Learners and Facilitating Learners Digital Competence showed a moderate and statistically significant correlation ($r = 0.491$, $p = 0.004$). This implies that educators who gained greater confidence in the assistance of the learners also had a greater capacity to steer the digital skill development of the students. This result is consistent with the expectation outlined in H3 and corresponds with other studies that learning instruction competence tends to intensify with the digital decision-making capacity of teachers (Langset et al., 2018; Lin & Kee, 2024). Although correlation does not imply direction, the analysis provides

Table 4 Correlation matrix in light of study objectives and hypotheses

Variable		Digital resources	Teaching and learning	Assessment	Empowering learners	Facilitating learners' digital competence
Digital Resources	Pearson's r	—				
	p-value	—				
Teaching and Learning	Pearson's r	−0.28	—			
	p-value	0.121	—			
Assessment	Pearson's r	−0.038	0.1	—		
	p-value	0.835	0.587	—		
Empowering Learners	Pearson's r	−0.254	0.129	−0.153	—	
	p-value	0.161	0.48	0.404	—	
Facilitating Learners' Digital Competence	Pearson's r	−0.091	0.054	−0.17	0.491	—
	p-value	0.619	0.77	0.353	0.004	—

Source: Author(s)

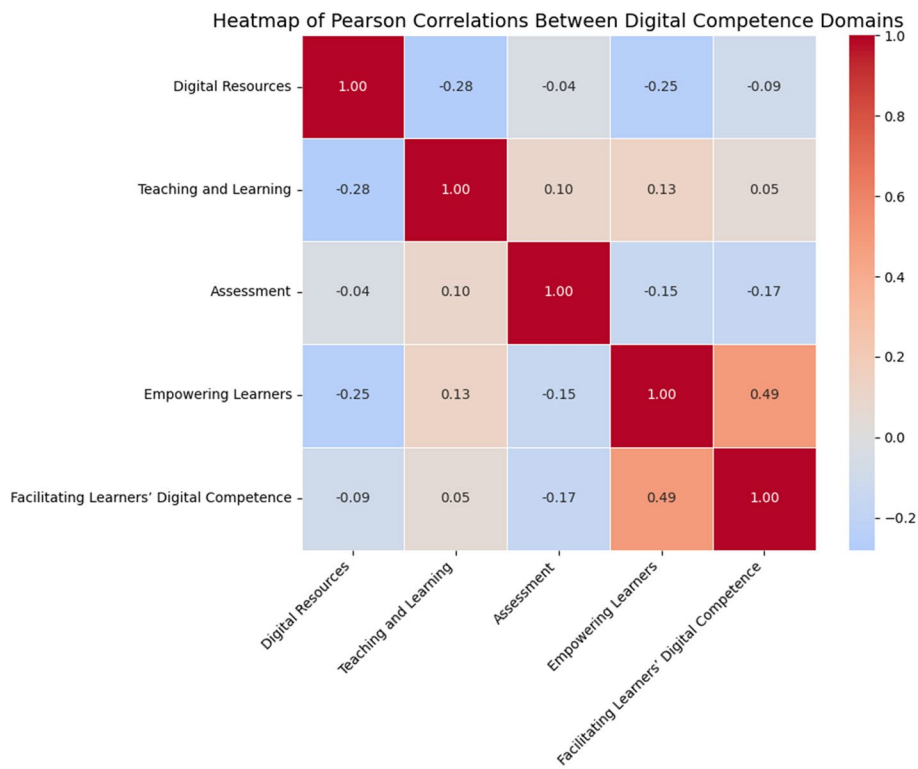

Fig. 6 Heatmap of Pearson correlations between digital competence domains. Source: Author(s)

exploratory evidence in support of H3, as statistically significant relationships were observed between learner empowerment and decision-making competencies following the intervention.

These relationships are illustrated in Fig. 6 in a graphical form, portraying the strength of correlations between domains. All other correlation values, including those for Digital Resources and Teaching and Learning ($r = -0.28$, $p = 0.121$), were weak or insignificant. This implies that although there was an improvement in all domains of the teachers, the development of these competencies was not necessarily linear or highly dependent on one another. According to previous research, domain-specific digital skills tend to develop autonomously and can rely on the focus of the PD modules and their nature (Jo Napitupulu et al., 2025; Tondeur et al., 2016).

The overall low or close to zero correlations (e.g., Assessment and Digital Resources: $r = -0.038$) can also be explained by the modular design of SmartPD, which covers each competency with specific microlearning units rather than addressing it in cross-domain tasks. This structure has enabled teachers to gain independence in each area, which is indicative of the intervention's intention to allow them to build discrete skill areas at their own pace.

In general, the patterns of correlation indicate that SmartPD supports not only a meaningful domain-specific enhancement but also flexible and self-directed learning, which is a focus of AI-based professional development models. These results contribute to a larger community of debates on scalable digital competence development in

education, making it possible for SmartPD to be implemented in diverse institutional settings (UNESCO, 2020).

4.5 Results for RQ2: teacher engagement and satisfaction with SmartPD

The qualitative analysis found three predominant themes connected with the teacher engagement and satisfaction with SmartPD, including (i) perceived usefulness of AI-generated feedback, (ii) flexibility and relevance of micro-learning design and (iii) collaborative reflection and peer learning. High behavioural engagement was achieved through xAPI analytics, which indicated that the rates of module completions and motivation to engage in any micro-learning activity were high. The qualitative reflection revealed that the satisfaction level was high, as teachers appreciated the timeliness and usefulness of AI-generated feedback, the relevance and flexibility of micro-lessons, and the opportunities for peer reflection. Overall, quantitative indicators and thematic insights demonstrate that teachers found SmartPD to be an interesting, convenient, and relevant tool in their context.

The quantitative findings were supported by qualitative data obtained through open-ended reflections, a short focus group discussion, and observation notes, which demonstrated the positive experience of teachers with the SmartPD model. Participants also expressed greater confidence in selecting and using digital tools. They mentioned that the micro-learning modules were well-organised and provided clear and realistic advice on how to use them in the classroom. The AI-generated feedback was valued by many, who described it as timely, specific, and helpful in identifying areas for improvement, which enhanced their self-regulated learning. The collaborative components of SmartPD were also appreciated by the teachers in the form of peer review and discussion, as they provided an opportunity for teachers to exchange ideas and see good examples of digital practices. All in all, these qualitative observations ensured that SmartPD was deemed not only approachable but also engaging and able to facilitate professional development, as the strong improvements observed throughout the quantitative data attest to. These qualitative insights complement the quantitative indicators of engagement, providing a deeper understanding of why the teachers were satisfied and engaged with the SmartPD intervention.

5 Discussion

Results of this study are consistent with and enhance previous DBR-based professional development interventions. Previous studies on DBR have produced incremental changes in the ICT integration practices of teachers, with practice-based models (Wang et al., 2014; Zinger et al., 2017). Nonetheless, such studies usually did not have powerful personalisation mechanisms, and they used more extended periods of intervention. In comparison, with the use of adaptive technologies and embedding AI-driven feedback loops and modular microlearning activities, SmartPD achieved significant, quantifiable changes within a relatively brief two-week cycle, which proved that DBR could succeed under these conditions.

Theoretically, the findings support the use of DBR as a sound model for connecting design, enactment, and evaluation in teacher professional development (McKenney & Reeves, 2018). SmartPD makes its contribution to this literature by demonstrating that

micro-iterations or daily refinement on prompts and AI-based feedback are valid refinements to DBR even in brief interventions. In practice, the results support the thesis that teacher PD has to go far beyond generic ICT training and adopt the context-sensitive, data-driven, and scalable models (Philipsen et al., 2019). SmartPD can demonstrate to policymakers and institutions the potential of modular, AI-assisted PD to integrate into the current professional learning systems in order to support teacher competence and the overall equity objectives of SDG 4.

Building on these broader insights from the literature, the present study specifically examined the design, implementation, and outcomes of the SmartPD model within a short two-week intervention. This study aimed to design, implement, and test the SmartPD model, which would help improve teachers' digital competencies and increase their digital learning with the help of microlearning and AI-based feedback (Liyanage et.al 2024; Tan et al., 2024)). The intervention was rooted in Design-Based Research (DBR) and lasted two weeks, with evaluation conducted by 32 teachers (Philipsen et al., 2019). The results of the effectiveness analysis, implementation, and correlation provide direct evidence indicating that SmartPD successfully met its main objectives.

To begin with, the successful implementation of the SmartPD model demonstrated its practical reality and educational significance (Philipsen et al., 2019). The users were able to navigate the system, and there were constant revisions of the modules to ensure they were responsive to the users. The interaction of microlearning and real-time performance tracking added a flexible and individualised experience to professional development (Ifenthaler & Schumacher, 2016; Viberg et al., 2018). The relevance of both the model usability and adapting to the context was also confirmed during formative observations during the intervention, satisfying Objective 1.

Second, with Objective 2 and Hypothesis H1, one can observe statistically significant changes ($p<.001$) in every one of the five areas of digital competence, or digital content creation, digital evaluation and diagnosis, digital instructional decision-making, and digital multimedia content production (Nguyen & Habók, 2023; Redecker, 2017). These findings demonstrate that the SmartPD model was effective in developing the participants' skills. The calculated pre-post mean differences suggest a considerable practical influence, while the decreased variability of post-test marks indicates improved uniformity of competence levels among the participants. This implies that SmartPD was effective not only for existing advanced users but also for those with lower entry skills.

Third, in line with Objective 3 and Hypothesis H2, it was noted that the post-intervention feedback showed good interest and satisfaction. Users enjoyed the experience of adaptive learning, as well as the relevance and ease of use of the content (Holmes et al., 2023; Wang & Han, 2022). The AI-mediated feedback provided the opportunity to offer insights promptly and promote the process of reflection and active learning. This adds to a growing body of evidence supporting the idea that individualised, modular professional learning tools have the potential to be a powerful stimulus for teacher motivation and learning (Tan et al., 2024).

On Hypothesis H3, there was a significant weak positive correlation between the Empowering Learners and Facilitating the Digital Competence of Learners ($r=0.491$, $p=0.004$). This observation suggests that teachers' confidence in empowering students through digital practices is a factor contributing to their ease in enabling students to

enhance their digital capabilities (Dang et al., 2024). The fact that the rest of the variables have low or non-significant correlations suggests that, although accompanied by improvements in the other variables, the improvement in digital competencies has not been dependent but rather happened in parallel with other variables, most likely due to the modular structure of the SmartPD framework.

Altogether, the SmartPD intervention indicated not only statistical but also pedagogical significance. It successfully contributed to the teacher's digital competence, facilitated reflective practice, and demonstrated the potential of AI and microlearning in specific professional development. The findings provide a reasonable basis for further scaling and contextualising the model, especially where learning experiences are required to be both accessible and personalised, particularly at the level of teacher learning (Stavermann, 2024; Khor & K, 2024). The correlational results can only be viewed as associative, rather than predictive, due to the small sample size and the exploratory nature of the analysis.

6 Practical implications

This study reveals that SmartPD is practically feasible as a disruptive and scalable model for teacher professional development. Its modular, micro-learning format, facilitated by AI-mediated formative feedback, allows for differentiated learning in addition to promoting digital competence, confidence, and reducing cognitive load among teachers (Hallmen et al., 2025b; Tan et al., 2024). These outputs of SmartPD can be directly translated into classroom practice by teachers through short H5P micro-lessons as digital warm-ups before the lesson, integrating AI-related formative assessments, and using proven AI recommendations to optimise lesson plans. Formalised routines of peer reflection, such as brief weekly vignette videos with rubrics to evaluate the videos, can be used to keep teachers engaged in reflective practice and collaborate to improve. There is also an institutional level, where xAPI analytics can be seen as a systematic way of tracking engagement, recognising potentially valuable teachers who could benefit from targeted coaching, and integrating SmartPD into long-term capacity-building plans (Stavermann, 2024; Viberg et al., 2018). The model can also be applied to environments with limited resources, utilising text-based or offline micro-modules to enhance accessibility and inclusivity. SmartPD also promotes equal access to high-quality professional learning, in line with SDG 4, enabling teachers to implement video-based digital assignments and data-driven scaffolding that can enhance learners' learning performance (Holmes et al., 2023; Philipsen et al., 2019). Due to its adaptable nature and contextual structure, SmartPD can be meaningfully integrated across diverse educational levels and fields, offering sustainable, real-time feedback loops that enhance not only instructional decision-making but also active pedagogical practice ((Khor & K, 2024; Stavermann, 2024).

7 Limitation

The study is associated with several limitations, despite the SmartPD intervention yielding positive results. Since the sample size is minimal (n = 32), the study may not yield generalizability, and future research should incorporate larger and more diverse groups of people. A reliance on digital infrastructure, including reliable internet access and suitable equipment, can limit the model's effectiveness in disadvantaged communities.

Additionally, the use of self-reported results introduces the possibility of response bias and the inability to determine long-term retention and classroom usage of competencies due to the short-term evaluation. Lastly, the results may have been biased by the participants' previous experience with digital tools, which requires that future implementations consider baseline differences.

Moreover, the study employed a limited use of Design-Based Research (DBR), focusing on a single short intervention cycle. Even though there were micro-iterations based on daily prompts and AI-based feedback refinements, as well as a series of redesign cycles in various contexts, the primary focus of the entire DBR methodology was not applied to the model (McKenney & Reeves, 2018; Wang & Hannafin, 2005). This restricts the degree to which iterative enhancements and situational changes can be formalised. Future efforts should focus on expanding SmartPD to multi-cycle DBR studies in various institutional and cultural contexts, allowing for a better understanding of scalability, sustainability, and long-term effects.

8 Conclusion

The current work aligns with the growing research on technology-enhanced teacher improvement. It develops the SmartPD model, an AI-supported, micro-learning-based model grounded in the DigCompEdu framework, created using a Design-Based Research (DBR) paradigm. The results indicate that short-cycle DBR interventions, combined with adaptive technologies and real-time formative feedback, can yield significant benefits to teachers' digital competence within a short timeframe. Notably, the research also elucidates that the competency relationship post-interventions must be understood as associative, rather than predictive, given the exploratory nature of the correlation analysis and the relatively modest sample size used; thus, maintaining methodological transparency in accordance with existing research standards.

Considering the future, there are a few promising directions for future development. To further confirm the adaptability and scalability of SmartPD, multi-cycle DBR investigations are required in varied institutional and cultural environments. Second, the longitudinal study would contribute to establishing whether the immediate benefits of increasing digital competence can be achieved through long-term improvements in teaching and, ultimately, by the learning outcomes among students. Third, the combination of immersive and emerging technologies, including voice feedback based on AI, augmented reality (AR), and virtual reality (VR), may also be considered in future work to customise a learning path further and enhance teacher involvement. Furthermore, the practical classroom examples presented in this study, including the use of micro-lessons, AI-based reflective routines, and xAPI-based analytics, demonstrate that SmartPD can be utilised to reinforce daily instructional decision-making. Lastly, by integrating SmartPD into institutional PD systems and national teacher development platforms, fair and high-quality professional learning opportunities can be promoted, which will add value to the goals of SDG 4: Quality Education.

Authors' contributions

SM led the conceptualization and design of the SmartPD model, developed the research framework, carried out the implementation, collected the data, conducted quantitative and qualitative analyses, and prepared the initial manuscript

draft. AR supervised the research design and methodological alignment, provided critical revisions to the analytical procedures and interpretations, contributed to the refinement of the manuscript for academic coherence, and acted as the corresponding author overseeing the final review and submission process. Both authors read and approved the final manuscript.

Funding
No funding was received to assist with the preparation of this manuscript.

Data availability
The analysis files and de-identified data from this study can be requested by the relevant author on reasonable grounds. None of the personally identifiable information is disclosed, and all results from the datasets align with institutional data protection policies.

Declarations

Ethics approval and consent to participate
Given in Manuscript Methodology section.

Consent for publication
Not applicable.

Competing interests
The authors have no conflicts of interest to declare that are relevant to the content of this article.

Received: 1 August 2025 Accepted: 22 January 2026
Published online: 05 February 2026

References

Ajani, O. A. (2024). Teachers' competencies in digital integration of learning contents in dynamic classroom practices: A review of teacher professional development needs. *Acta Educationis Generalis, 14*(3), 18–40. https://doi.org/10.2478/atd-2024-0016

Albion, P. R., Tondeur, J., Forkosh-Baruch, A., & Peeraer, J. (2015). Teachers' professional development for ICT integration: Towards a reciprocal relationship between research and practice. *Education and Information Technologies, 20*(4), 655–673. https://doi.org/10.1007/s10639-015-9401-9

Alcaraz-Domínguez, S., & Barberà, E. (2025). Developing educators' digital competences through microlearning: Analysis of training needs in Spain. In Proceedings of the 2024 16th International Conference on Education Technology and Computers (ICETC '24) (pp. 423–428). Association for Computing Machinery. https://doi.org/10.1145/3702163.3702450

Banihashem, S. K., Noroozi, O., Khosravi, H., Schunn, C. D., & Drachsler, H. (2025). Pedagogical framework for hybrid intelligent feedback. Innovations in Education and Teaching International, 1–17. https://doi.org/10.1080/14703297.2025.2499174

Betancur-Chicué, V., & García-Valcárcel Muñoz-Repiso, A. (2023). Microlearning for the development of teachers' digital competence related to feedback and decision making. *Education Sciences, 13*(7), Article 722. https://doi.org/10.3390/educsci13070722

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology, 3*(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Cross, S., Wolfenden, F., Charania, A., Adinolfi, L., Sen, S., & Sarkar, D. (2022). SUPPORTING PRACTICE-BASED TEACHER PROFESSIONAL LEARNING AND ASSESSMENT AT SCALE IN THE GLOBAL SOUTH. INTED Proceedings, 1, 5431–5440. https://doi.org/10.21125/inted.2022.1404

Dang, T. D., Phan, T. T., Vu, T. N. Q., La, T. D., & Pham, V. K. (2024). Digital competence of lecturers and its impact on student learning value in higher education. *Heliyon, 10*(17), e37318. https://doi.org/10.1016/j.heliyon.2024.e37318

ELM Learning. (2024, June 5). XAPI. https://elmlearning.com/hub/technology-tools/xapi/

Elsayary, A. (2023). The impact of a professional upskilling training programme on developing teachers' digital competence. *Journal of Computer Assisted Learning, 39*, 1154–1166. https://doi.org/10.1111/jcal.12788

Fakhar, H., Lamrabet, M., Echantoufi, N., Khattabi, K. E., & Ajana, L. (2024). Towards a new artificial intelligence-based framework for teachers' online continuous professional development programs: Systematic review. International Journal of Advanced Computer Science and Applications, 15(4). https://doi.org/10.14569/ijacsa.2024.0150450

Fernández-Batanero, J. M., Montenegro-Rueda, M., Fernández-Cerero, J., & García-Martínez, I. (2020). Digital competences for teacher professional development. Systematic review. *European Journal of Teacher Education, 45*(4), 513–531. https://doi.org/10.1080/02619768.2020.1827389

Fowler, S., & Leonard, S. N. (2021). Using design-based research to shift perspectives: A model for sustainable professional development for the innovative use of digital tools. *Professional Development in Education, 50*, 192–204. https://doi.org/10.1080/19415257.2021.1955732

Frasier PY, Slatt L, Kowlowitz V, Kollisch DO, Mintzer M. Focus groups: a useful tool for curriculum evaluation. Fam Med. 1997 Jul-Aug;29(7):500–7. PMID: 9232413. https://pubmed.ncbi.nlm.nih.gov/9232413

Fredricks, J. A., Blumenfeld, P. C., & Paris, A. H. (2004). School Engagement: The Potential of the Concept and State of the Evidence. *Review of Educational Research, 74*(1), 59–109. https://doi.org/10.3102/00346543074001059

Gisbert Cervera, M., & Lázaro Cantabrana, J. L. (2015). Professional development in teacher digital competence and improving school quality from the teachers' perspective: A case study. *Journal of New Approaches in Educational Research, 4*, 115–122. https://doi.org/10.7821/naer.2015.7.123

Hallmen, T., Gietl, K., Hillesheim, K., Bauermann, M., Friedrich, A., & André, E. (2025b). AI-Based feedback in counselling competence training of prospective teachers. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2505.03423

Henderson, M., Ryan, T., & Phillips, M. (2019). The challenges of feedback in higher education. *Assessment & Evaluation in Higher Education, 44*(8), 1237–1252. https://doi.org/10.1080/02602938.2019.1599815

Holmes, W., Bialik, M., & Fadel, C. (2023). Artificial Intelligence in Education: Promises and Implications for Teaching and Learning. OECD Publishing. https://www.oecd.org/education/ceri/AI-in-Education.pdf

Ifenthaler, D., & Schumacher, C. (2016). Student perceptions of privacy principles for learning analytics. *Educational Technology Research and Development, 64*(5), 923–938. https://doi.org/10.1007/s11423-016-9477-y

International Society for Technology in Education. (2022). ISTE standards for educators. ISTE. https://www.iste.org/standards/educators

Kara, M. (2020). Distance education: A systems view of online learning: By Michael G. Moore and Greg Kearsley, Belmont, CA, Wadsworth Cengage Learning, 2012, 361 pp., £92.99 (paperback), ISBN: 978-1-111-52099-1. *Educational Review, 72*(6), 800. https://doi.org/10.1080/00131911.2020.1766204

Khor, E. T., & K, M. (2024). A Systematic Review of the Role of Learning Analytics in Supporting Personalized Learning. Education Sciences, 14(1), 51. https://doi.org/10.3390/educsci14010051

Koh, J. H. L., & Chai, C. S. (2016). Seven design frames that teachers use when considering technological pedagogical content knowledge (TPACK). *Computers & Education, 102*, 244–257. https://doi.org/10.1016/j.compedu.2016.09.003

Langset, I. D., Jacobsen, D. Y., & Haugsbakken, H. (2018). Digital professional development: Towards acollaborative learning approach for taking higher education intothe digitalized age. *Nordic Journal of Digital Literacy, 13*(1), 24–39. https://doi.org/10.18261/issn.1891-943x-2018-01-03

Liyanage, N G L S J & Sandamali, K.A.D. & Fernando, Purnima & Sisindra, K. (2024). THE NEED AND EFFECTIVENESS OF MICROLEARNING IN TEACHER EDUCATION: A SYSTEMATIC LITERATURE REVIEW. https://ours.ou.ac.lk/wp-content/uploads/2024/12/ID-107_IRC-OUSL-2024.pdf

McKenney, S., & Reeves, T. C. (2018). *Conducting Educational Design Research.* https://doi.org/10.4324/9781315105642

Napitupulu, M. H., Muddin, A., Bagiya, N., Diana, S., & Rosyidah, N. S. (2025). Teacher Professional Development in the Digital Age: Strategies for Integrating technology and Pedagogy. International Journal for Science Review, 2(4). https://doi.org/10.71364/ijfsr.v2i4.33

Nguyen, La. T., & Habók, A. (2023). Tools for assessing teacher digital literacy: A review. *Journal of Computers in Education, 11*(1), 305–346. https://doi.org/10.1007/s40692-022-00257-5

Párraga, L. M., Cejudo, C. L., & Osuna, J. B. (2022). Validation of the DigCompEdu check in questionnaire through structural equations: A study at a university in Peru. *Education Sciences, 12*(8), Article 574. https://doi.org/10.3390/educsci12080574

Philipsen, B., Tondeur, J., Pareja Roblin, N., et al. (2019). Improving teacher professional development for online and blended learning: A systematic meta-aggregative review. *Educational Technology Research and Development, 67*, 1145–1174. https://doi.org/10.1007/s11423-019-09645-8

Punie, Y., editor(s), Redecker, C. (2017). European Framework for the Digital Competence of Educators: DigCompEdu , EUR 28775 EN, Publications Office of the European Union, Luxembourg, ISBN 978–92–79–73718–3 (print),978–92–79–73494–6 (pdf). https://doi.org/10.2760/178382. (print), 10.2760/159770 (online) JRC107466

Schürmann, V., Bodemer, D., & Marquardt, N. (2025). Exploring the use of regular reflections in student collaboration: A case study in higher education. *Frontiers in Education*. https://doi.org/10.3389/feduc.2025.1526487

Sharathbabu, Vemula. (2025). Professional Development for Digital Literacy: Insights from Secondary Data on Teacher Training Programs. https://www.researchgate.net/publication/397738545_Professional_Development_for_Digital_Literacy_Insights_from_Secondary_Data_on_Teacher_Training_Programs

Shavelson, R. J. (1973). What is the basic teaching skill? *Journal of Teacher Education, 24*(2), 144–151. https://doi.org/10.1177/002248717302400213

Stavermann, K. (2024). Online teacher professional development: A research synthesis on effectiveness and evaluation. *Technology Knowledge and Learning, 30*(1), 203–240. https://doi.org/10.1007/s10758-024-09792-9

Syifa, N. S. N., Az-Zahra, N. a. M., & Rachman, N. I. F. (2024). Analisis Infrastruktur Teknologi, Pelatihan Pengajar dan Tantangan dalam Implementasi Model Pembelajaran Literasi Digital untuk Mendukung SDGs 2030. Jurnal Sadewa Publikasi Ilmu Pendidikan Pembelajaran Dan Ilmu Sosial, 2(2), 212–224. https://doi.org/10.61132/sadewa.v2i2.817

Tan, X., Cheng, G., & Ling, M. H. (2024). Artificial intelligence in teaching and teacher professional development: A systematic review. Computers and Education: Artificial Intelligence, 8, Article 100355. https://doi.org/10.1016/j.caeai.2024.100355

Jo Tondeur, Alona Forkosh-Baruch, Sarah Prestridge, Peter Albion, & Shiyama Edirisinghe. (2016). Responding to Challenges in Teacher Professional Development for ICT Integration in Education. Journal of Educational Technology & Society, 19(3), 110–120. http://www.jstor.org/stable/jeductechsoci.19.3.110

Tregubova, T., Shibankova, L. A., & Kats, A. S. (2020). Conceptual framework for teachers' professional development design in the era of digitalization. *ARPHA Proceedings, 1*, 2595–2608. https://doi.org/10.3897/ap.2.e2595

UNESCO. (2018). ICT Competency Framework for Teachers (Version 3). United Nations Educational, Scientific and Cultural Organisation. https://unesdoc.unesco.org/ark:/48223/pf0000265721

UNESCO. (2020). Education for sustainable development: A roadmap. United Nations Educational, Scientific and Cultural Organisation. https://unesdoc.unesco.org/ark:/48223/pf0000374802

UNESCO. (2023). Global education monitoring report 2023: Technology in education – A tool on whose terms? United Nations Educational, Scientific and Cultural Organisation. https://unesdoc.unesco.org/ark:/48223/pf0000385723

Viberg, O., Hatakka, M., Bälter, O., & Mavroudi, A. (2018). The current landscape of learning analytics in higher education. *Computers in Human Behavior, 89*, 98–110. https://doi.org/10.1016/j.chb.2018.07.027

Wang, Z., & Han, F. (2022). The effects of teacher feedback and automated feedback on cognitive and psychological aspects of foreign language writing: A mixed-methods research. *Frontiers in Psychology, 13*, Article 909802. https://doi.org/10.3389/fpsyg.2022.909802

Wang, F., & Hannafin, M. J. (2005). Design-based research and technology-enhanced learning environments. *Educational Technology Research and Development, 53*(4), 5–23. https://doi.org/10.1007/BF02504682

Wuryaningsih, W., Darwin, M., Susilastuti, D.H., & Pierewan, A.C. (2018). Online learning as an innovative model of Teachers' Professional Development (TPD) in the digital era: A literature review. Character Education for 21st Century Global Citizens. https://doi.org/10.1201/9781315104188-9

Zinger, D., Naranjo, A., Amador, I. F., Gilbertson, N., & Warschauer, M. (2017). A design-based research approach to improving professional development and teacher knowledge: The case of the Smithsonian Learning Lab. Contemporary Issues in Technology and Teacher Education, 17(3), 388–410. https://citejournal.org/volume-17/issue-3-17/general/a-design-based-research-approach-to-improving-professional-development-and-teacher-knowledge-the-case-of-the-smithsonian-learning-lab/

Wang, F., Hsu, H. Y., Reeves, T. C., & Coster, D. C. (2014). Professional development to enhance teachers' practices in using information and communication technologies (ICTs) as cognitive tools: Lessons learned from a design-based research study. *Computers & Education, 79*, 101–115. https://doi.org/10.1016/j.compedu.2014.07.006

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.